

FINAL MOCK

Answer all the questions below

(Note: You will get two coding problems and one tracing problem in your real final exam.)

(1)

Write a Python function that takes an integer number as input from the user. Without converting the number to a string, find the reverse of the string of the number, but keep only the even digits, and also find the sum of all the even numbers.

Input	Output
4356	Reverse: 64 Sum: 10
12345984	Reverse: 4842 Sum: 18

(2)

Assume you are given two dictionaries. The dictionary named `current_sem_data` contains GPA of 3 students of their current semester courses. Another dictionary named `prev_sem_data` contains their previous semesters CGPA and credits completed respectively. Assume all courses are of 3 credit points each.

```
current_sem_data = {  
    "Messi": (4.0, 3.7, 4.0),  
    "Ronaldo": (3.3, 3.7, 4.0),  
    "Neymar": (3.3, 3.7, 4.0)  
}
```

```
prev_sem_data = {  
    "Messi": (3.88, 90),  
    "Ronaldo": (3.52, 70),  
    "Neymar": (3.25, 85)  
}
```

Now write a python programme that will take a student's name as user input and calculate and print that student's current CGPA using both dictionaries.

Formulas:

$\text{current_total_GPA} = \sum \text{current_course_grade} * \text{credit}$

$\text{previous_total_GPA} = \text{previous_CGPA} * \text{credits_completed}$

$\text{CGPA} = (\text{current_total_GPA} + \text{previous_total_GPA}) / \text{Total_credits_completed}$

Input	Output
Neymar	####Data for Neymar#### Prev Sem Credit: 85 Prev Sem CGPA: 3.25 Credits Completed with Current Sem: 94 Current Sem GPA: 33.0 CGPA: 3.29

(3)

```
players = {
    "Benzema": {
        "Ovr": 99,
        "Pos": "CF",
        "Pac": 97,
        "Sho": 99,
        "Pas": 93,
    },
    "Lewandoski": {
        "Ovr": 97,
        "Pos": "ST",
        "Pac": 90,
        "Sho": 99,
        "Pas": 86,
    },
    "Tonali": {
        "Ovr": 92,
        "Pos": "CDM",
        "Pac": 90,
        "Sho": 85,
        "Pas": 91,
    }
}
```

You are given a player's dictionary named `players` that contains some player information from FIFA Ultimate Team (Fifa 23). Your task is to write a Python function that must take the player name as a parameter and have different attributes. Then you have to create a new dictionary with the information about that player and his attributes and finally return the new dictionary. If no attribute is entered, return a dictionary with the player name and all the attributes.

Input	Return
func("Tonali")	{'Tonali': {'Ovr': 92, 'Pos': 'CDM', 'Pac': 90, 'Sho': 85, 'Pas': 91}}
func("Benzema", "Pos", "Sho")	{'Benzema': {'Pos': 'CF', 'Sho': 99}}
func("Lewandoski", "Pac", "Pos", "Pas")	{'Lewandoski': {'Pac': 90, 'Pos': 'ST', 'Pas': 86}}